

Requirements Document — AI-Enhanced Writing Web Application

Requirement 1

Title: User Registration

Description: Allow new users to create an account.

Input: Email, password, display name

Processing: Validate email format, hash password, store user profile

Output: Confirmation of successful registration

Error Handling: Notify user if email already exists or validation fails

Requirement 2

Title: User Login

Description: Authenticate returning users.

Input: Email, password

Processing: Verify credentials against stored records

Output: Access to user dashboard

Error Handling: Display incorrect credentials message

Requirement 3

Title: Story Project Creation

Description: Enable users to create new writing projects.

Input: Project title, optional description

Processing: Store project metadata

Output: New project added to user workspace

Error Handling: Notify user if project cannot be created

Requirement 4

Title: Generative-AI Name Generator

Description: Generate names for characters, places, or objects.

Input: Category (person/place/object), optional style tags

Processing: Generative-AI model produces names

Output: List of suggested names

Error Handling: Provide fallback generic names if AI fails

Requirement 5

Title: Generative-AI Storyline Idea Generator

Description: Provide creative prompts and plot ideas.

Input: Genre, tone, optional keywords

Processing: Generative-AI model creates storyline concepts

Output: 3–10 storyline ideas

Error Handling: Display message if generation fails

Requirement

Title: Generative-AI Image Creation Tool

Description: Create images for characters, settings, or scenes.

Input: Text prompt, style preferences

Processing: AI image model renders image

Output: Downloadable/generated image

Error Handling: Provide message if image generation fails

Requirement 7

Title: AI-Chatbot Writing Assistant

Description: Provide conversational writing help.

Input: User messages, story context

Processing: AI chatbot generates responses

Output: Writing suggestions, edits, or explanations

Error Handling: Notify user if chatbot is unavailable

Requirement 8

Title: AI-Translation Tool for Multilingual Text

Description: Translate text for multilingual characters or settings.

Input: Source text, target language

Processing: AI translation model processes text

Output: Translated text

Error Handling: Provide fallback translation or error message

Requirement 9

Title: AI-Translation for a Grammar and Style Checker

Description: Improve clarity and correctness of writing.

Input: User-submitted text

Processing: AI analyzes grammar, tone, and style

Output: Suggested corrections and improvements

Error Handling: Notify user if analysis fails

Requirement 10

Title: Generative-AI Story Outline Builder

Description: Generate structured outlines for stories.

Input: Genre, theme, story length

Processing: Generative-AI model creates multi-section outline

Output: Outline document

Error Handling: Provide simplified outline if AI fails

Requirement 11

Title: Generative-AI Character Profile Generator

Description: Create detailed character bios.

Input: Character role, genre, traits

Processing: Generative-AI model generates profile

Output: Character description

Error Handling: Provide minimal template if AI fails

Requirement 12

Title: Generative-AI World-Building Assistant

Description: Generate world details (cultures, geography, lore).

Input: Genre, world type

Processing: Generative-AI model produces world-building elements

Output: World description

Error Handling: Provide fallback generic world template

Requirement 13

Title: User Feedback Submission

Description: Allow users to submit feedback or requests for admins to view.

Input: Feedback text, category

Processing: Store feedback and notify admin

Output: Confirmation message

Error Handling: Notify user if submission fails

Requirement 14

Title: AI-Assisted Admin Dashboard (AI Admin Assistant)

Description: Help admins manage user requests using AI.

Input: User feedback, support tickets

Processing: AI categorizes and prioritizes requests

Output: Sorted and labeled admin queue

Error Handling: Default to manual sorting

Requirement 15

Title: AI-Recommendation Engine for Shared Ideas

Description: Suggest ideas created by other users.

Input: User preferences, browsing history

Processing: AI ranks and recommends relevant ideas

Output: Personalized recommendation list

Error Handling: Show generic popular ideas

Requirement 16

Title: Collaborative Writing Workspace

Description: Allow multiple users to collaborate on a story.

Input: Edits, comments

Processing: Sync changes in real time

Output: Updated shared document

Error Handling: Notify users of sync issues

Requirement 17

Title: Version History Tracking

Description: Track changes to writing projects.

Input: Document edits

Processing: Store version snapshots

Output: Version list with timestamps

Error Handling: Warn user if version cannot be saved

Requirement 18

Title: Secure File Storage

Description: Store user documents and images securely.

Input: Uploaded files

Processing: Encrypt and store files

Output: File accessible in project

Error Handling: Notify user if upload fails

Requirement 19

Title: Search Functionality

Description: Allow users to search across projects and ideas.

Input: Search query

Processing: Index lookup and ranking

Output: Search results

Error Handling: Display no-results message

Requirement 20

Title: Notification System

Description: Notify users about updates, recommendations, or AI outputs.

Input: Trigger events

Processing: Generate and send notifications

Output: In-app or email notification

Error Handling: Log failed notifications

Requirement 21

Title: User Profile Management

Description: Allow users to update personal information.

Input: Profile fields

Processing: Validate and save updates

Output: Updated profile

Error Handling: Notify user if update fails

Requirement 22

Title: AI-Powered Writing Metrics Analyzer

Description: Provide analytics on writing habits.

Input: User text, activity logs

Processing: AI analyzes patterns

Output: Metrics dashboard

Error Handling: Display partial metrics if analysis fails

Requirement 23

Title: AI-Powered Content Moderation System

Description: Ensure user-generated content meets guidelines.

Input: Text or images

Processing: AI flags inappropriate content

Output: Approval or moderation notice

Error Handling: Escalate to human review

Requirement 24

Title: Export Tools

Description: Allow users to export stories in various formats.

Input: Selected format (PDF, TXT, etc.)

Processing: Convert and package content

Output: Downloadable file

Error Handling: Notify user if export fails

Requirement 25

Title: Accessibility Features

Description: Provide accessible UI options.

Input: User accessibility settings

Processing: Apply UI adjustments

Output: Customized interface

Error Handling: Revert to default settings if configuration fails

Requirement 26

Title: AI-Assisted User Behavior Monitoring

Description: Track and analyze user activity so that admins can detect unusual or harmful behavior.

Input: User activity logs, interaction patterns

Processing: AI identifies anomalies or potential misuse

Output: Alerts or flagged accounts for admin review

Error Handling: Default to manual review if AI detection fails

Requirement 27

Title: Admin Content Review Queue

Description: Provide admins with a dashboard to review AI-flagged content.

Input: Flagged posts, images, comments

Processing: Sort and categorize items by severity

Output: Review list with action options

Error Handling: Notify admin if queue fails to load

Requirement 28

Title: Admin Role & Permission Management

Description: Allow super-admins to assign roles and permissions to admins.

Input: Admin account, selected permissions

Processing: Update access control lists

Output: Updated admin privileges

Error Handling: Revert changes if permission update fails

Requirement 29

Title: System Health & AI-Model Monitoring

Description: Monitor AI model performance, latency, and error rates.

Input: Model logs, performance metrics

Processing: AI analyzes trends and identifies issues

Output: Health dashboard with alerts

Error Handling: Fallback to basic system logs

Requirement 30

Title: Admin Audit Log Viewer

Description: Track all actions done by admins for accountability.

Input: Admin activity logs

Processing: Organize logs chronologically

Output: Searchable audit trail

Error Handling: Notify admin if logs cannot be retrieved

Requirement 31

Title: AI-Powered Support Ticket Auto-Responder

Description: Provide automated responses to common support requests to ease the support workload for admins.

Input: User support ticket

Processing: AI identifies ticket type and drafts response

Output: Suggested or auto-sent reply

Error Handling: Route ticket to human admin if AI fails

Requirement 32

Title: Admin Announcement & Broadcast Tool

Description: Allow admins to send platform-wide announcements.

Input: Announcement text, target audience

Processing: Format and schedule broadcast

Output: In-app or email announcement

Error Handling: Log failed broadcasts

Requirement 33

Title: Admin Data Export & Reporting

Description: Allow admins to export platform usage data.

Input: Date range, report type

Processing: Compile analytics and generate report

Output: CSV, PDF, or dashboard view

Error Handling: Notify admin if export fails

Requirement 34

Title: AI-Driven Trend Analysis for User Content

Description: Identify trending genres, themes, or tools.

Input: User content metadata

Processing: AI analyzes patterns and clusters

Output: Trend insights for admins

Error Handling: Provide partial analysis if AI fails